InletTracker Compatibility Changelog

Purpose: This document tracks all modifications made to the original InletTracker/CoastSat codebase to make it run on a modern stack (Windows 11, Python 3.10, and current versions of GDAL, Earth Engine API, NumPy, scikit-image, Shapely, and pandas as of 2026). The original codebase was written around 2020–2021 against much older library versions, and most of these fixes address breaking changes introduced by newer releases of those libraries — not bugs in the original science/logic.

Environment used:

- OS: Windows 11
- Conda environment: Python 3.10
- Key packages: gdal=3.6.4, numpy=1.26.4, earthengine-api=0.1.370, spyder=5.4.3, scikit-image (0.25+), current shapely (2.x), current pandas (2.x+)

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1. Google Earth Engine / Download Pipeline (coastsat/SDS_download.py)

Issue	**Fix**
Landsat Collection 1 asset paths (LANDSAT/LC08/C01/T1_TOA, etc.) no longer exist in Earth Engine — GEE migrated to Collection 2.	Updated all col_names_T1 (and T2) dictionary entries from C01 to C02.
ee.Initialize() failed with "no project found" — GEE now requires an explicit Cloud project tied to Earth Engine registration.	Changed to ee.Initialize(project='your-project-id').
gdal_merge import (import gdal_merge) no longer resolves — it moved into the osgeo_utils namespace in modern GDAL packaging.	Changed to from osgeo_utils import gdal_merge.
Version-check logic int(ee.__version__[-3:]) broke because modern earthengine-api version strings (e.g. 1.7.32) don't parse as a 3-digit integer.	Fixed version parsing logic in download_tif().
Several while True: try: ... except: continue loops silently retried forever on **any** exception,	Changed bare except: to except Exception as e: print(e) (at least during

masking real errors (e.g. bad collection paths) as infinite hangs with no error message.	debugging) so real failures surface instead of looping silently.

2. Image Preprocessing (coastsat/SDS_preprocess.py)

Issue	**Fix**
Downloaded band data sometimes contained -inf/inf values, which corrupted downstream matplotlib rendering (blank image windows) and broke percentile/rescale calculations.	Added np.where(np.isinf(...), np.nan, ...) and negative-value cleanup on im_ms after preprocess_single().
exposure.rescale_intensity() in rescale_image_intensity() crashed (IndexError: index -1 is out of bounds) when an entire band was masked/NaN (empty array passed to np.percentile).	Added a check for zero-length valid-pixel arrays before calling np.percentile/rescale_intensity; skips rescale for fully-masked bands instead of crashing.
morphology.remove_small_objects(... , in_place=True) — the in_place keyword argument was removed in scikit-image 0.25. This appeared **twice** in create_cloud_mask() (once in the base path, once inside the cloud_mask_issue branch), and the second instance only triggered once an image actually needed that code path — which is why it appeared to "start failing" partway through a run.	Removed in_place=True and reassigned the return value instead: cloud_mask = morphology.remove_small_objects(cloud_mask , min_size=..., connectivity=1) (both instances).
morphology.square(3) is deprecated in scikit-image 0.25+ (removed in 0.27).	Flagged for future replacement with morphology.footprint_rectangle((3,3)) — not yet mandatory, currently just a FutureWarning.
DataFrame.append() (pandas, if present in this file) removed in pandas 2.0.	Checked — no DataFrame-style .append() calls found in this file as currently modified; only

	plain Python list.append() calls remain, which are unaffected.

3. Coordinate/Projection Handling (coastsat/SDS_tools.py)

Issue	Fix
Persistent ERROR 1: PROJ: utm: Invalid latitude / Reprojection failed, err = 2049 messages during preprocessing. Root cause: the convert_epsg() function was passing coordinates to PROJ in the wrong axis order for the PROJ/GDAL version in use (GDAL's axis-order handling changed between major versions — this is the classic "lat/lon vs lon/lat" GDAL 3.x pitfall). This silently corrupted pixel bounding-box coordinates, which downstream surfaced as bounding boxes of -9223372036854775808 (the min-int64 error sentinel) in get_bounding_box_minmax(), causing displayed images to be cropped completely out of view and causing **100% of images across all satellites and sites to fail pathfinding** ("no path could be found") in Step 4.	Corrected axis order handling in convert_epsg() (fix contributed via a second AI debugging pass after this conversation's summary was handed off — see note below). This was the single highest-impact fix in the whole porting effort.

4. Training Data / Display (inlettracker/InletTracker_tools.py)

Issue	Fix
Blank/invisible image windows during Step 2 (create_training_data) manual classification, even though image arrays themselves were valid in QGIS.	Traced to the same PROJ/bounding-box corruption above; once convert_epsg was fixed, bounding boxes returned valid pixel coordinates and images displayed correctly. As a defensive measure, also added a fallback in set_openvsclosed(): if Xmin/Xmax/Ymin/Ymax are clearly invalid (e.g. large negative sentinel values), fall back to displaying the full image extent instead of an invalid crop.

-inf/NaN band values also needed cleanup here (mirrors the SDS_preprocess.py fix) before display.	Added the same np.isinf / negative-value NaN cleanup after preprocess_single() calls in create_training_data().

5. Automated Pathfinding (Step 4, inlettracker/InletTracker_tools.py)

Issue	**Fix**
intersection = intersection[0] — indexing a Shapely geometry directly to grab the first point of a multi-part intersection result. This worked in Shapely 1.x (where a MultiPoint was subscriptable) but raises TypeError in Shapely 2.x. The surrounding bare except: swallowed this error silently, leaving intersection as an un-indexed MultiPoint/MultiLineString, which then crashed later with NotImplementedError when .coords was accessed on a multi-part geometry. This only manifested on images where the along-berm and across-berm transects crossed **more than once** — explaining why some images worked and others didn't.	Replaced the indexing with Shapely 2.x-correct handling: check intersection.geom_type, pull the first sub-geometry via .geoms for MultiPoint/MultiLineString/GeometryCollection results, and handle the LineString (collinear overlap) case explicitly by taking its first vertex.
gdf_all = gdf_all.append(gdf) (and similar) — DataFrame.append()/GeoDataFrame.append() was removed entirely in pandas 2.0.	Replaced all instances with gdf_all = pd.concat([gdf_all, gdf]) (several locations across InletTracker_tools.py, Step 4 and Step 5 post-processing).
Cloud cover over the inlet-specific ROI was reporting a suspicious, constant value (0.407) for every single L8 image regardless of actual cloud content, while the whole-image cloud cover varied normally. Root cause: the Collection 2 QA band mask used for cloud_mask_issue likely includes water/shadow-adjacent bits (e.g. dilated	**Partially addressed / flagged for follow-up:** recommended restricting the L8 QA bitmask check to just bits 3 (cloud) and 2 (cirrus) rather than including shadow/dilated-cloud bits, which should bring the "clear day" floor down to ~0.000

cloud / cloud shadow bits) that happen to flag water pixels near the inlet channel — exactly the pixels the pathfinder needs.	(matching S2 behavior). Not fully verified/closed out at time of writing.

6. Post-Processing / Time Series (Step 5, inlettracker/InletTracker_tools.py)

Issue	**Fix**
KeyError: Value based partial slicing on non-monotonic DatetimeIndexes with non-existing keys is not allowed in subset_DTM_df_in_time(). Root cause: the dataframe mixes L8 and S2 rows in **processing order**, not chronological order. Older pandas silently allowed df[startdate:enddate] slicing on an unsorted DatetimeIndex; modern pandas refuses it outright.	Added Input_df = Input_df.sort_index() immediately after the first datetime index is built (before the [startdate:enddate] slice), *not* in the later string-reformatting index block further down the function — placement matters here.
ChainedAssignmentError / SettingWithCopyWarning on lines using the df['col'][mask] = value pattern (in the threshold-optimization loop and in classify_image_series_via_DTM). Currently only a warning (assignments still work under Copy-on-Write today), but will silently stop working under pandas 3.0.	Replaced with .loc chained assignment: df.loc[mask, 'col'] = value.

7. Environment / Tooling Notes (not code changes, but relevant context)

- **scikit-image version drift mid-project**: the in_place removal (see §2) was not actually caused by a version *upgrade* happening mid-session — it was a code path (cloud_mask_issue=True) that simply hadn't been exercised yet by earlier images. Worth remembering: an error "appearing out of nowhere" after previously-successful runs may mean a rarer code branch is being hit for the first time, not that the environment changed.
- **Windows + OneDrive**: shapefile write permission errors (Layer creation failed: permission denied) were traced to working inside a OneDrive-synced folder; saving

outputs outside OneDrive (e.g. directly under Documents or Desktop, not the OneDrive-synced version) avoids intermittent lock/permission issues.

- **QGIS display of .tif outputs**: raw multispectral GeoTIFFs will appear as blank/noisy images in standard Windows image viewers — they must be opened in a GIS tool like QGIS, which understands nodata values and band scaling.

Summary

In total, porting this ~2021-era codebase involved fixing breaking changes across **six major dependencies**: GDAL/OGR (Collection 2 migration, axis order), the Earth Engine Python API (auth/project requirements, version parsing), NumPy/scikit-image (removed in_place kwarg, deprecated square), Shapely (1.x→2.x multi-geometry indexing), and pandas (.append() removal, chained-assignment/Copy-on-Write, non-monotonic datetime slicing). None of these fixes change the underlying scientific method — they restore the original intended behavior on a modern software stack.

Document prepared based on a full debugging session porting InletTracker to a 2026 Windows/conda environment, covering two test sites (Carmel and Pajaro) for the 2019 calendar year, with Pajaro carried through to full Step 5 post-processing and validation.